

PROJECT SYNOPSIS

ON

NVIDIA BROADBAND BILLING SYSTEM

Submitted in Partial Fulfillment of the Requirements for the Degree of

MASTER OF COMPUTER APPLICATION (MCA)

Academic Year: 2026–2027

Submitted By:

MCA – II, Semester III

Under the Guidance of:

Submitted To:

1. INTRODUCTION

The broadband and Internet Service Provider (ISP) industry requires efficient management of customers, broadband plans, billing, payments, data usage and service requests. In traditional broadband service environments, many operational activities are performed manually or through partially computerized systems. Customer registration, plan allocation, bill generation, payment collection and complaint handling may involve paperwork, spreadsheets or disconnected systems.

Such conventional processes can result in duplicate records, incorrect data entry, delayed payment updates, billing errors and difficulty in tracking customer complaints. Customers may also face difficulties in viewing their billing history, checking outstanding amounts and obtaining information about their broadband services.

The **NVDIA Broadband Billing System** is proposed as a centralized web-based application to automate and simplify broadband service management. The system provides facilities for customer registration and login, broadband plan management, billing, payment tracking, data usage monitoring, service request management and announcements.

The proposed system aims to reduce manual work, improve billing accuracy, provide better transparency and enhance the overall efficiency of broadband service operations.

2. PROBLEM STATEMENT

Existing broadband service management systems may depend on manual records, spreadsheets and separate processes for customer management, billing, payments and complaints. These approaches make it difficult to efficiently manage a large volume of customer information.

Manual billing can lead to errors in bill amounts, due dates and late fee calculations. Payment information may not be updated immediately, causing confusion regarding payment status and outstanding balances. Customers may also have limited access to previous bills and transaction information.

Similarly, complaints and service requests may be reported through telephone calls or personal visits without a structured tracking mechanism. This makes it difficult

for administrators to monitor requests, assign priorities and ensure timely resolution.

Therefore, there is a need for a centralized, secure and automated broadband billing and management system that integrates customer management, broadband plans, billing, payment tracking, data usage and service requests into a single platform.

3. AIM OF THE PROJECT

The main aim of the project is to develop a centralized web-based Broadband Billing and Management System that automates broadband service operations and provides efficient management of customers, plans, billing, payments, data usage and service requests.

4. OBJECTIVES OF THE PROJECT

The major objectives of the proposed system are:

1. To develop an automated and centralized Broadband Billing and Management System.
2. To provide secure customer registration and login facilities.
3. To allow customers to view and subscribe to available broadband plans.
4. To provide an efficient module for monthly bill generation and management.
5. To allow customers to view bill amounts, due dates, late fees and payment status.
6. To maintain payment and transaction history.
7. To provide data usage tracking for monitoring internet consumption.
8. To provide a service request module for raising broadband-related complaints.
9. To enable administrators to manage and resolve customer service requests.
10. To provide an announcement module for publishing important updates and notifications.

5. SCOPE OF THE PROJECT

The scope of the **NVDIA Broadband Billing System** covers the major activities involved in broadband service management.

5.1 Customer Management

The system allows customers to register, create an account and securely log in to access broadband services.

5.2 Broadband Plan Management

Customers can view available broadband plans and select a suitable plan based on speed, validity and pricing. Administrators can add, update and remove broadband plans.

5.3 Billing Management

The system supports bill generation and management. Customers can view their bill amount, due date, late fee and payment status.

5.4 Payment and Transaction Management

The system maintains payment status and transaction records, allowing customers and administrators to track payment-related information.

5.5 Data Usage Management

Customers can monitor their internet consumption and view their usage information.

5.6 Service Request Management

Customers can raise complaints or service requests and upload attachments where required. Administrators can view and manage these requests and update their status.

5.7 Announcement Management

Administrators can publish important announcements such as offers, maintenance notifications and policy-related updates.

6. PROPOSED SYSTEM

The proposed system is a web-based application consisting of a frontend, backend and relational database.

The frontend is developed using React.js, while the backend is implemented using Java and Spring Boot. Spring MVC is used for REST API development, Spring Security and JWT are used for authentication and authorization, and Spring Data JPA with Hibernate provides database interaction. MySQL is used as the primary relational database.

The frontend communicates with the backend through REST APIs. The backend processes requests, performs business operations and communicates with the MySQL database. This architecture provides modularity, maintainability and efficient communication between system components.

7. DATABASE DESIGN

The system uses **MySQL** as its primary relational database.

The major database tables identified in the project are:

Sr. No.	Table	Purpose
1	admin_sign_in	Stores administrator authentication details
2	broadband_plans	Stores broadband plan information
3	bills	Stores billing and payment records
4	announcements	Stores system announcements
5	data_usage	Stores internet usage information
6	service_requests	Stores customer service requests

Spring Data JPA and Hibernate are used for Object Relational Mapping between Java entity classes and database tables.

10. FUNCTIONAL REQUIREMENTS

The system shall provide the following functional requirements:

1. The system shall allow customers to register.
2. The system shall allow registered customers to log in.
3. The system shall provide administrator login functionality.
4. The administrator shall be able to manage broadband plans.
5. Customers shall be able to view available broadband plans.
6. The system shall support bill generation.
7. Customers shall be able to view billing information.
8. The system shall maintain payment status and transaction records.
9. The system shall support data usage tracking.
10. Customers shall be able to raise service requests.
11. Administrators shall be able to view and manage service requests.
12. Administrators shall be able to publish announcements.
13. The system shall store application information in MySQL.
14. The frontend and backend shall communicate through REST APIs.
15. The system shall support CRUD operations for relevant modules.

11. NON-FUNCTIONAL REQUIREMENTS

11.1 Security

The system uses authentication and authorization mechanisms, including JWT-based authentication and Spring Security.

11.2 Reliability

The system should maintain consistent customer, billing and service-request information.

11.3 Performance

The system should provide efficient responses during normal operations and repeated API requests.

11.4 Maintainability

The layered backend architecture allows individual components to be maintained and modified independently.

11.5 Scalability

The system architecture allows future features and additional modules to be integrated.

11.6 Usability

The system provides a user-friendly interface for both customers and administrators.

12. TECHNOLOGY REQUIREMENTS

12.1 Software Requirements

Technology	Purpose
Java 17	Backend programming
Spring Boot	Backend framework
Spring MVC	REST API development
Spring Security	Authentication and authorization
JWT	Token-based authentication
Spring Data JPA	Database interaction
Hibernate	ORM
MySQL	Relational database
Maven	Dependency and build management
React.js	Frontend development
OpenPDF	PDF bill generation

These technologies are specified in the project report's technology stack.

12.2 Hardware Requirements

The project requires a computer system capable of running the development environment, Java/Spring Boot backend, React.js frontend and MySQL database.

Minimum practical requirements include:

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB or above
- Storage: 20 GB or above available space
- Network: Internet/LAN connectivity

15. EXPECTED OUTCOMES

The proposed system is expected to:

1. Reduce manual work involved in broadband service management.
2. Improve accuracy of customer and billing records.
3. Provide centralized customer and service information.
4. Simplify broadband plan management.
5. Provide efficient bill generation and tracking.
6. Improve payment and transaction record management.
7. Allow customers to monitor data usage.
8. Provide structured complaint and service request management.
9. Improve communication through announcements.
10. Provide secure access to system functionality.
11. Improve transparency between customers and the broadband service provider.
12. Improve overall operational efficiency and customer satisfaction.

16. FUTURE ENHANCEMENTS

The system can be further enhanced with:

- Integration of secure online payment gateways.
- OTP-based authentication and password recovery.
- Email and SMS notifications.
- PDF invoice generation and download.
- Automatic monthly bill generation.
- Analytics dashboard for revenue, users and plan usage.
- Mobile application using React Native or Flutter.
- Live chat or chatbot support.
- Broadband plan upgrade/downgrade functionality.
- Real-time data usage monitoring.
- Priority levels for service requests.
- Multi-language support.
- Advanced role-based access control.
- Database backup and recovery.
- Audit logs for administrator activities.

17. CONCLUSION

The **NVDIA Broadband Billing System** is designed as a centralized web-based solution for managing broadband service operations. It integrates customer registration, broadband plan management, billing, payment tracking, data usage monitoring, service requests and announcements into a single platform.

The system uses **React.js for the frontend, Java/Spring Boot for the backend and MySQL for database management**, with Hibernate and Spring Data JPA supporting database interaction. The layered architecture helps provide modularity and maintainability.

The proposed system addresses major limitations of manual broadband service management by reducing paperwork, improving billing accuracy, centralizing information and providing better customer support. The completed project testing reported successful execution of all critical test scenarios, demonstrating the functional feasibility of the system.